

EXPL 118 - 13

End-to-End Systems Security

Fall 2026

Course Summary

This course is intended to give you the experience of working in an active research environment, and prepare you for future contributions in our area of research. For first time students (AKA first cohort), much of the quarter will be spent skill-building and absorbing the [Rust](#) programming language, operating systems, security, and more! If you're taking this course for the second time (AKA second cohort), welcome back! You will be taking the skills you learned the first time around and applying them towards active research directions in the lab. Additionally, everyone will participate in weekly paper readings in the field.

Learning Outcomes

By the end of this quarter you should be able to...

- Utilize Rust at a comfortable enough level to use it for your own projects.
- Read and digest papers in the field.
- Contribute towards current research directions within the lab.
- Gain research skills that stick with you beyond your participation in the lab.

Course Staff

The best way to reach out to course staff is going to be through the [LSD Lab Zulip chat](#). Please reach out to us via email if you have not been added yet.

Name	Role	Email
Owen Arden	Instructor	oarden@ucsc.edu
Surendra Jammishetti	TA	sjammish@ucsc.edu
Max Carr	TA	mcarr35@ucsc.edu

Course Logistics

Most asynchronous communication will happen through the LSD lab Zulip (but we may send some announcements both Zulip and Canvas).

Grading

Category	Weight
Attendance	10%
Mid Quarter Notebook Review	5%
Final Notebook Review	15%
Assignments	10%
Final Report	20%

Final Presentation	15%
Peer Evaluations	10%
Active Contribution	15%

Attendance

Attendance will be taken every class session. You are allowed 2 unexcused absences but if something comes up, please get in touch with course staff before your absence.

Lab Notebooks

All students are expected to maintain a **Research Notebook** to document their weekly progress, things they find interesting, to-do items etc.

Your notebook must be an instantiation of this [template repository](#).

Mid Quarter Notebook Review

During week 5, you will be expected to submit your notebook to a TA, in the form of a GitHub repository link. This review is intended to be a check-in to help you make sure you're working on your notebook properly.

Final Notebook Review

You will submit your final notebook the same way as the mid quarter review. The course staff will evaluate it as a whole and assign it a grade. Note that if we requested improvements during your Mid Quarter review, and the notebook entries since then incorporate those improvements, you will not be penalized for the first half of the notebook!

Lab Assignments (First Cohort)

The first cohort will have lab assignments related to the Rust language, as well as getting things running inside the Twizzler OS. Lab due dates will be posted once labs are released. We will do our best to ensure labs are able to be completed in the allotted time, but if not, we may consider extending the due dates.

We do not allow AI use on lab assignments as their purpose is to work with and internalize the material – read more in the [AI policy](#) section. Nevertheless, feel free to discuss lab assignments with peers during class hours, although all submitted work must be your own. **We trust that you know the line between copy pasting and discussing ideas.**

Lab assignments are considered submitted when successfully checked off by a TA anytime before the lab due date.

You are also encouraged to ask TAs for help regarding these assignments.

Research Project (Second Cohort)

Instead of lab assignments, the second cohort will each be working on a research thrust of their choice. In the first few weeks of the course you will set long-term goals, as well as weekly actionable items, in collaboration with the teaching staff. The rest of the quarter will consist of making reasonable progress towards those goals. The expectation is to complete your long-term goal by the end of the quarter. However, we know first-hand that it's not always possible, so you can always discuss with the course staff to revise what you think can be accomplished by the end of the quarter.

Your final grade in this category will be decided by how well you executed your goal, lessons learned, any resulting artifacts, etc. This will vary per project so work closely with the course staff!

Grading scale

This will be the grading scale used when assigning final grades to percentages. There may also be a class bonus for filling out SETs, announced later in the quarter.

Grade	Min	Max
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A+	97%	100%
A	93%	97%
A-	90%	93%
B+	87%	90%
B	83%	87%
B-	80%	83%
C+	77%	80%
C	70%	77%
D+	67%	70%
D	63%	67%
D-	60%	63%
F	0%	60%

Academic Honesty / AI Policy

All work you submit must be your own, but as this is a research class, discussion between peers is encouraged. As stated earlier, **we trust you know the difference between discussion and copy-pasting**.

We also are aware that AI tools are quite useful, but can also take away from your learning of the material (especially when trying to learn a new programming language). For much of the class, feel free to responsibly use AI tools but usage is **prohibited** in the following activities, since we have designed them to be the most helpful when completed without AI assistance:

- Any and all writing work, including all parts of your notebook.
- Assignments.

If you find yourself needing help with any aspect of the class, feel free to reach out to your peers and course staff first!